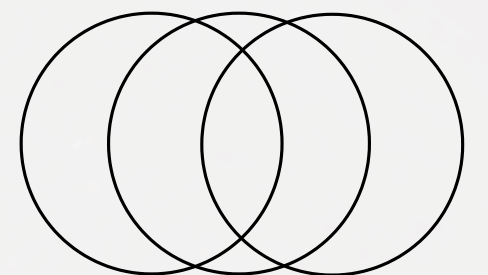


SQL PROJECT 1

JAKARTA AIR POLLUTION TREND ANALYSIS (2010 - 2025)

QORI ALFIANI HAKIKI



• PROJECT OVERVIEW

This project focuses on analyzing **Jakarta's air quality** data using **SQL** as the primary tool for data processing and analysis.

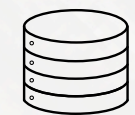
• DATA

The data consists of **two separate tables**, with a total of **5,537 records**.



dim_stasiun.csv

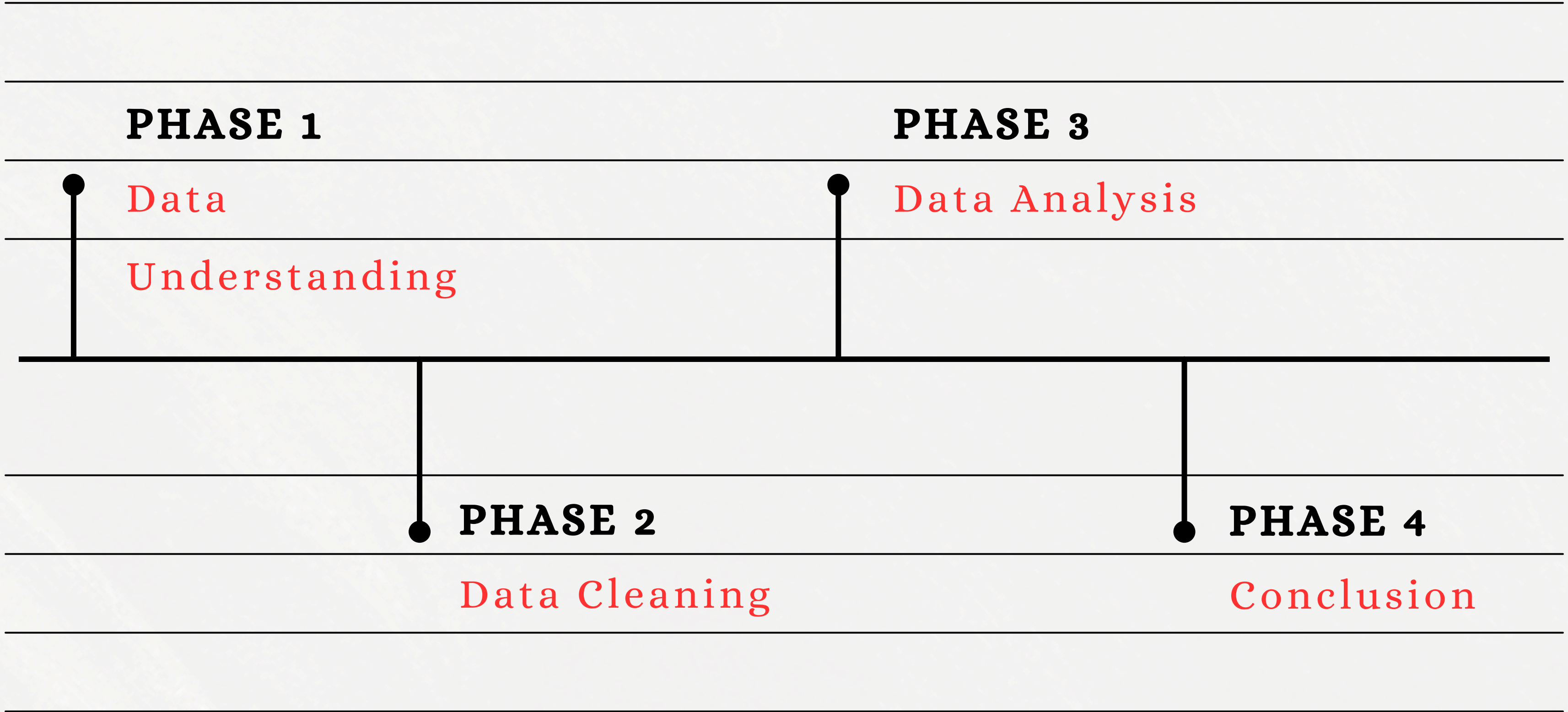
stasiun_id	nama_stasiun
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fact_ispu.csv

stasiun_id	tanggal	pm10	pm25	co	o3	categori
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• **WORKFLOW**



- **PHASE 1: DATA UNDERSTANDING**

- nama_stasiun contains 5 distinct monitoring stations.

- Time-series data from 2010 to 2025.

- tanggal is not in date format.

- Columns PM10, PM2.5, CO, and O₃ represent daily pollutant concentrations.

stats	pm10	pm25	co	o3
minimum	2	10	1	4
maximum	187	287	119	314

- **PHASE 1: DATA UNDERSTANDING**

- 4,205 null values across PM10, PM2.5, CO, and O₃, as shown below.

pm10_null	pm25_null	co_null	o3_null
315	4022	89	104

- categori represents air quality levels based on ISPU standards, determined by the dominant pollutant.

• PHASE 2: DATA CLEANING

- Before data cleaning, I created a temporary table named `ispu_analysis` by joining the `dim_stasiun` and `fact_ispu` tables using `stasiun_id` as the key.

- No duplicate data was found in the `ispu_analysis` table.

- The `tanggal` column data type was standardized to `DATE`.

```
CREATE TEMPORARY TABLE ispu_analysis AS
SELECT
    fi.stasiun_id,
    fi.tanggal,
    fi.pm10, fi.pm25, fi.co, fi.o3,
    fi.categori,
    ds.nama_stasiun
FROM fact_ispu fi
JOIN dim_stasiun ds
ON fi.stasiun_id = ds.stasiun_id;
```


- **PHASE 2: DATA CLEANING**

- Two patterns were identified to handle null values:
-

- Pattern 1: 4 rows with the category “TIDAK ADA DATA” contain all null values, so they were replaced with 0.
-

- Pattern 2: each category follows a specific pollutant range:
-

- BAIK: 0-50

- SEDANG: 51-100

- TIDAK SEHAT: 101-200

- SANGAT TIDAK SEHAT: 201-300

- BERBAHAYA: >300

• **PHASE 2: DATA CLEANING**

• To verify my assumption,	Kategori	Status Warna	Angka Rentang
I searched for references	Baik	Hijau	1-50
and found the ISPU	Sedang	Biru	51-100
standard, which defines	Tidak Sehat	Kuning	101-200
pollutant ranges for each	Sangat Tidak Sehat	Merah	201-300
	Berbahaya	Hitam	≥301

Source: <https://jurnal.untan.ac.id/index.php/jcskommipa/article/view/58704>

category.

• Data was filtered based on these conditions, resulting in 5,435 valid records (including null values) and 102 inconsistent records.

- **PHASE 2: DATA CLEANING**

```
WHERE kategori = 'BAIK' AND (  
    (pm10 BETWEEN 0 AND 50 OR pm10 IS NULL) AND  
    (pm25 BETWEEN 0 AND 50 OR pm25 IS NULL) AND  
    (co BETWEEN 0 AND 50 OR co IS NULL) AND  
    (o3 BETWEEN 0 AND 50 OR o3 IS NULL)
```

- Examples of valid and inconsistent records are shown in the table below.

pm10	pm25	co	o3	kategori	
27	50	12	30	BAIK	VALID
60	NULL	73	27	SEDANG	VALID
52	78	15	11	SEDANG	INVALID

- **PHASE 2: DATA CLEANING**

- Instead of deleting inconsistency data, a new validated table (ispu_clean) was created containing only consistency data, resulting in a total of 5,435 records.

- After identifying consistent patterns and creating a clean dataset, null values were imputed using the median of each category. For example, rows in the “BAIK” category were filled using the median within the 0–50 range.

• PHASE 2: DATA CLEANING

```
UPDATE ispu_clean AS target
SET target.pm10 = (
  SELECT AVG(pm10_val)
  FROM (
    SELECT pm10 AS pm10_val,
           ROW_NUMBER() OVER (ORDER BY pm10) AS rn,
           COUNT(*) OVER () AS total_rows
    FROM ispu_clean
    WHERE kategori = 'BAIK' AND
           pm10 IS NOT NULL
  ) AS temp
  WHERE rn IN (
    FLOOR((total_rows + 1)/2),
    FLOOR((total_rows + 2)/2)
  )
)
WHERE target.pm10 IS NULL
AND target.kategori = 'BAIK';
```

	stasiun_id	tanggal	pm10	pm25	co	o3	kategori	nama_stasiun
▶	1	2019-07-07	63	80	12	41	SEDANG	DKI1 (Bunderan HI)
	1	2020-06-10	67	80	10	34	SEDANG	DKI1 (Bunderan HI)
	1	2020-10-09	60	80	7	87	SEDANG	DKI1 (Bunderan HI)
	1	2020-10-10	53	80	7	60	SEDANG	DKI1 (Bunderan HI)
	1	2020-12-09	32	45	5	25	BAIK	DKI1 (Bunderan HI)
	1	2020-12-10	32	45	4	25	BAIK	DKI1 (Bunderan HI)
	1	2021-12-21	52	74	19	22	SEDANG	DKI1 (Bunderan HI)
	1	2022-02-06	39	51	15	57	SEDANG	DKI1 (Bunderan HI)
	1	2022-09-13	59	81	18	45	SEDANG	DKI1 (Bunderan HI)
	1	2022-10-07	51	66	17	35	SEDANG	DKI1 (Bunderan HI)
	1	2022-10-18	59	79	16	45	SEDANG	DKI1 (Bunderan HI)
	1	2023-03-01	38	44	8	19	BAIK	DKI1 (Bunderan HI)
	1	2023-03-02	29	33	11	21	BAIK	DKI1 (Bunderan HI)
	1	2023-03-31	45	57	13	22	SEDANG	DKI1 (Bunderan HI)
	1	2023-04-11	27	51	12	14	SEDANG	DKI1 (Bunderan HI)
	1	2023-04-15	65	94	12	27	SEDANG	DKI1 (Bunderan HI)

• **PHASE 3: DATA ANALYSIS**

• After applying filtering based	<pre>SELECT CONCAT(MIN(pm10), '-', MAX(pm10)) AS pm10_range, CONCAT(MIN(pm25), '-', MAX(pm25)) AS pm25_range, CONCAT(MIN(co), '-', MAX(co)) AS co_range, CONCAT(MIN(o3), '-', MAX(o3)) AS o3_range, kategori FROM ispu_clean GROUP BY kategori</pre>
on ISPU standards, the table	
below shows that each category	
follows a distinct pollutant	

range.

	pm10_range	pm25_range	co_range	o3_range	kategori	
►	0-0	0-0	0-0	0-0	TIDAK ADA DATA	
	5-50	26-50	2-50	4-50	BAIK	
	2-100	25-100	1-99	4-100	SEDANG	
	5-187	58-199	1-119	6-199	TIDAK SEHAT	
	18-163	217-287	3-69	23-297	SANGAT TIDAK SEHAT	
	73-73	80-80	28-28	314-314	BERBAHAYA	

• **PHASE 3: DATA ANALYSIS**

• Checked the number of records per year to ensure consistency. The data shows that 2025 contains only 59 days, while 2010–2024 have full-year records. • Then to observe the trend, the average frequency of each air quality category per year (2010–2024) was calculated to understand how pollution levels change over time.		COUNT(*)	YEAR(tanggal)
	▶	365	2010
		365	2011
		364	2012
		365	2013
		363	2014
		365	2015
		366	2016
		349	2017
		355	2018
		365	2019
		347	2020
		365	2021
		360	2022
		348	2023
		334	2024
		59	2025

• PHASE 3: DATA ANALYSIS

Average frequency per year

	tahun	baik	sedang	tidak_sehat	sangat_tidak_sehat	berbahaya
►	2010	0.2411	0.6904	0.0493	0.0110	0.0000
	2011	0.0822	0.5041	0.4000	0.0137	0.0000
	2012	0.0082	0.2692	0.4066	0.3132	0.0027
	2013	0.0110	0.4192	0.4959	0.0740	0.0000
	2014	0.0331	0.7025	0.2479	0.0165	0.0000
	2015	0.1178	0.7068	0.1753	0.0000	0.0000
	2016	0.0710	0.6721	0.2541	0.0027	0.0000
	2017	0.1232	0.5616	0.3152	0.0000	0.0000
	2018	0.0310	0.3662	0.5268	0.0761	0.0000
	2019	0.0055	0.4795	0.4932	0.0219	0.0000
	2020	0.0519	0.7118	0.2277	0.0086	0.0000
	2021	0.0082	0.6110	0.3808	0.0000	0.0000
	2022	0.0083	0.6111	0.3806	0.0000	0.0000
	2023	0.0172	0.5057	0.4655	0.0086	0.0000
	2024	0.0419	0.7006	0.2575	0.0000	0.0000

• Air quality levels show **no consistent increase or decrease** from 2010 to 2024. Notably, even during the COVID-19 period (2019–2023), there was no significant improvement in air quality.

• **PHASE 3: DATA ANALYSIS**

Worst vs best quality				<pre>YEAR(tanggal) AS tahun, ROUND(AVG(categori = 'BAIK'),2) AS baik, ROUND(AVG(categori = 'SEDANG'),2) AS sedang, ROUND(AVG(categori = 'TIDAK SEHAT') + AVG(categori = 'SANGAT TIDAK SEHAT') + AVG(categori = 'BERBAHAYA'),2) AS tingkat_buruk</pre>	
tahun	baik	sedang	tingkat_buruk		
2012	0.01	0.27	0.72		
2018	0.03	0.37	0.60		
2013	0.01	0.42	0.57		
2019	0.01	0.48	0.52		
2023	0.02	0.51	0.47		
2011	0.08	0.50	0.41		
2021	0.01	0.61	0.38		• The worst year is based on the highest combined proportion of “Tidak Sehat”, “Sangat Tidak Sehat”, and “Berbahaya” categories, with 2012 as the worst and 2010 as the lowest (best).
2022	0.01	0.61	0.38		
2017	0.12	0.56	0.32		
2024	0.04	0.70	0.26		
2014	0.03	0.70	0.26		
2016	0.07	0.67	0.26		
2020	0.05	0.71	0.24		
2015	0.12	0.71	0.18		
2010	0.24	0.69	0.06		

• **PHASE 3: DATA ANALYSIS**

Weekday vs weekend

pm10	pm25	co	o3	categori	jenis_hari
73	80	28	314	BERBAHAYA	Weekend
77	249	33	232	SANGAT TIDAK SEHAT	Weekday
51	249	17	204	SANGAT TIDAK SEHAT	Weekend
65	249	31	207	SANGAT TIDAK SEHAT	Weekend
52	249	24	208	SANGAT TIDAK SEHAT	Weekday
62	249	17	210	SANGAT TIDAK SEHAT	Weekday
58	249	18	211	SANGAT TIDAK SEHAT	Weekday
66	249	38	212	SANGAT TIDAK SEHAT	Weekday
79	249	31	214	SANGAT TIDAK SEHAT	Weekend
69	249	23	215	SANGAT TIDAK SEHAT	Weekday
61	249	17	220	SANGAT TIDAK SEHAT	Weekday
66	249	24	236	SANGAT TIDAK SEHAT	Weekday
64	249	27	226	SANGAT TIDAK SEHAT	Weekday
72	249	24	226	SANGAT TIDAK SEHAT	Weekday
82	249	23	207	SANGAT TIDAK SEHAT	Weekend
60	249	14	213	SANGAT TIDAK SEHAT	Weekend
63	249	15	201	SANGAT TIDAK SEHAT	Weekday

• The distribution of air quality categories between weekdays and weekends was nearly identical. This indicates that pollution levels were relatively stable throughout the week.

• **PHASE 3: DATA ANALYSIS**

Dominan polutan

tahun	dominan_polutan	jumlah	<pre>YEAR(tanggal) AS tahun, CASE WHEN pm10 >= pm25 AND pm10 >= co AND pm10 >= o3 THEN 'PM10' WHEN pm25 >= pm10 AND pm25 >= co AND pm25 >= o3 THEN 'PM25' WHEN co >= pm10 AND co >= pm25 AND co >= o3 THEN 'CO' ELSE 'O3' END AS dominan_polutan</pre>
2010	PM25	302	
2011	PM25	152	
2012	O3	184	
2013	O3	203	
2014	PM25	167	• PM2.5 dominates most years, consistently showing the highest frequency compared to PM10, CO, and O₃.
2015	PM25	196	
2016	PM25	199	
2017	PM25	174	
2018	O3	216	
2019	O3	228	
2020	PM25	176	
2021	PM25	339	
2022	PM25	334	
2023	PM25	344	
2024	PM25	321	
2025	PM25	57	

• **PHASE 3: DATA ANALYSIS**

Pollution spike				<pre>SELECT tanggal, pm10, LAG(pm10) OVER(ORDER BY tanggal) AS hari_sebelum, pm10 - LAG(pm10) OVER(ORDER BY tanggal) AS lonjakan, kategori FROM ispu_clean</pre>	
tanggal	pm25	hari_sebelumnya	lonjakan		
2010-11-10	249	80	169		
2010-12-24	249	80	169		
2012-05-02	249	80	169		
2012-11-05	249	80	169		
2012-11-19	249	80	169		
2014-12-31	249	80	169		
2018-04-05	249	80	169		
2018-05-08	249	80	169		
2018-11-16	249	80	169		
2019-08-23	249	80	169		
2010-11-18	249	113	136		
2010-11-22	249	113	136		
2011-09-23	249	113	136		
2011-10-11	249	113	136		
2011-12-23	249	113	136		
2012-02-28	249	113	136		

- This analysis identifies sudden spikes in PM2.5 levels by **comparing each day's value with the previous day.**
- **Extreme daily spikes** (up to +169), in 2012, indicating frequent short-term pollution surges.

• PHASE 4: CONCLUSION

- The analysis confirms that air quality categories follow distinct pollutant ranges aligned with ISPU standards.
 - Yearly trends (2010–2024) show no clear improvement or deterioration over time, even during the COVID-19 period.
 - While 2010 represents the best air quality, 2019 reflects consistently poor conditions, and 2012 stands out as the most extreme year due to the presence of “BERBAHAYA” levels and frequent pollution spikes.
-

• PHASE 4: CONCLUSION

- air quality remains relatively stable between weekdays and weekends.

- PM_{2.5} is identified as the dominant pollutant across most years, with occasional extreme daily spikes (up to +169), indicating short-term pollution surges.

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The End

THANK YOU